Cybersecurity Hands-On Lab

Module 2: Network Discovery and Scanning with Nmap

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Lab Overview

This lab introduces students to network reconnaissance, the first phase of penetration testing. Students will use Nmap to discover devices, identify open ports, and analyze exposed services.

Part 1: Verify Installation

nmap –version: v 7.92

```
yitelpin@anoen-x:~$ nmap --version
Nmap version 7.92 ( https://nmap.org )
Platform: x86_64-redhat-linux-gnu
Compiled with: nmap-liblua-5.3.5 openssl-3.5.4 libssh2-1.11.1 libz-1.3.1.zlib-ng libpcre2-10.47 libpcap-1.10.6 nmap-libdnet-1.12 ipv6
Compiled without:
Available nsock engines: epoll poll select
yitelpin@anoen-x:~$
```

Part 2: Discover Active Hosts

Two active IP addresses:

1: 192.168.100.25

2: 192.168.100.107

```
yitelpin@anoen-x:~$ nmap -sn 192.168.100.0/24
Starting Nmap 7.92 ( https://nmap.org ) at 2026-03-24 11:48 GMT
Nmap scan report for _gateway (192.168.100.1)
Host is up (0.013s latency).
Nmap scan report for 192.168.100.25
Host is up (0.041s latency).
Nmap scan report for 192.168.100.52
Host is up (0.067s latency).
Nmap scan report for 192.168.100.59
Host is up (0.067s latency).
Nmap scan report for 192.168.100.106
Host is up (0.57s latency).
Nmap scan report for 192.168.100.107
Host is up (0.034s latency).
Nmap scan report for 192.168.100.133
Host is up (0.18s latency).
Nmap scan report for anoen-x (192.168.100.139)
Host is up (0.00023s latency).
Nmap scan report for 192.168.100.153
Host is up (0.41s latency).
Nmap scan report for 192.168.100.187
Host is up (0.14s latency).
Nmap scan report for 192.168.100.196
Host is up (0.13s latency).
Nmap scan report for 192.168.100.198
Host is up (0.13s latency).
Nmap scan report for 192.168.100.204
Host is up (0.14s latency).
Nmap scan report for 192.168.100.207
Host is up (1.2s latency).
Nmap scan report for 192.168.100.208
Host is up (0.11s latency).
Nmap scan report for 192.168.100.213
Host is up (0.21s latency).
Nmap scan report for 192.168.100.214
Host is up (0.13s latency).
Nmap scan report for 192.168.100.217
Host is up (0.13s latency).
Nmap scan report for 192.168.100.220
Host is up (1.3s latency).
Nmap scan report for 192.168.100.221
Host is up (0.62s latency).
```

Part 3: Scan Open Ports

List three open ports:

1: 135/tcp open

2: 139/tcp open

3: 445/tcp open

```
yitelpin@anoen-x:~$ nmap 192.168.100.107
Starting Nmap 7.92 ( https://nmap.org ) at 2026-03-24 12:04 GMT
Nmap scan report for 192.168.100.107
Host is up (0.23s latency).
Not shown: 997 closed tcp ports (conn-refused)
PORT    STATE SERVICE
135/tcp open  msrpc
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 2.10 seconds
yitelpin@anoen-x:~$
```

Part 4: Service Detection

One detected service and version:

----- 139/tcp open netbios-ssn Microsoft Windows netbios-ssn

```
yitelpin@anoen-x:~$ nmap -sV 192.168.100.107
Starting Nmap 7.92 ( https://nmap.org ) at 2026-03-24 12:09 GMT
Nmap scan report for 192.168.100.107
Host is up (0.028s latency).
Not shown: 997 closed tcp ports (conn-refused)
PORT    STATE SERVICE       VERSION
135/tcp open  msrpc         Microsoft Windows RPC
139/tcp open  netbios-ssn   Microsoft Windows netbios-ssn
445/tcp open  microsoft-ds?
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 14.59 seconds
yitelpin@anoen-x:~$
```

Part 5: OS Detection

Detected OS: OS: Windows; CPE: cpe:/o:microsoft:windows

```
yitelpin@anoen-x:~$ sudo nmap -O 192.168.100.107
Placez votre doigt sur le lecteur d'empreintes
Starting Nmap 7.92 ( https://nmap.org ) at 2026-03-24 12:12 GMT
Nmap scan report for 192.168.100.107
Host is up (0.031s latency).
Not shown: 997 closed tcp ports (reset)
PORT    STATE SERVICE
135/tcp open  msrpc
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds
MAC Address: 22:A2:11:F7:34:62 (Unknown)
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.92%E=4%D=3/24%OT=135%CT=1%CU=35563%PV=Y%DS=1%DC=D%G=Y%M=22A211%
OS:TM=69C27FAF%P=x86_64-redhat-linux-gnu)SEQ(SP=103%GCD=1%ISR=109%TI=I%CI=I
OS:%II=I%SS=S%TS=A)SEQ(SP=103%GCD=1%ISR=109%TI=I%CI=I%TS=A)OPS(O1=M5B4NW8ST
OS:11%O2=M5B4NW8ST11%O3=M5B4NW8NNT11%O4=M5B4NW8ST11%O5=M5B4NW8ST11%O6=M5B4S
OS:T11)WIN(W1=FFFF%W2=FFFF%W3=FFFF%W4=FFFF%W5=FFFF%W6=FFFF)ECN(R=Y%DF=Y%T=8
OS:0%W=FFFF%O=M5B4NW8NNS%CC=N%Q=)T1(R=Y%DF=Y%T=80%S=O%A=S+%F=AS%RD=0%Q=)T2(
OS:R=Y%DF=Y%T=80%W=0%S=Z%A=S%F=AR%O=%RD=0%Q=)T3(R=Y%DF=Y%T=80%W=0%S=Z%A=O%F
OS:=AR%O=%RD=0%Q=)T4(R=Y%DF=Y%T=80%W=0%S=A%A=O%F=R%O=%RD=0%Q=)T5(R=Y%DF=Y%T
OS:=80%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)T6(R=Y%DF=Y%T=80%W=0%S=A%A=O%F=R%O=%RD=
OS:0%Q=)T7(R=Y%DF=Y%T=80%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)U1(R=Y%DF=N%T=80%IPL=
OS:164%UN=0%RIPL=G%RID=G%RIPCK=G%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=80%CD=Z)

Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 14.73 seconds
```

Part 6: Aggressive Scan

```
yitelpin@anoen-x:~$ sudo nmap -A 192.168.100.107
Placez votre doigt sur le lecteur d'empreintes
Starting Nmap 7.92 ( https://nmap.org ) at 2026-03-24 12:27 GMT
Nmap scan report for 192.168.100.107
Host is up (0.067s latency).
Not shown: 997 closed tcp ports (reset)
PORT    STATE SERVICE       VERSION
135/tcp open  msrpc         Microsoft Windows RPC
139/tcp open  netbios-ssn   Microsoft Windows netbios-ssn
445/tcp open  microsoft-ds?
MAC Address: 22:A2:11:F7:34:62 (Unknown)
Aggressive OS guesses: Microsoft Windows 10 1703 (99%), Microsoft Windows 10 1507 - 1607 (97%), Microsoft Windows 10 1511 (95%), Microsoft Windows Longhorn (95%), Mic
rosoft Windows 7 SP0 - SP1, Windows Server 2008 SP1, Windows Server 2008 R2, Windows 8, or Windows 8.1 Update 1 (94%), Microsoft Windows Server 2008 SP2 (93%), Micros
oft Windows 8 (93%), Microsoft Windows 10 (93%), Microsoft Windows Vista SP1 (93%), Microsoft Windows 10 10586 - 14393 (93%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_clock-skew: -1s
| smb2-security-mode:
|   3.1.1:
|_    Message signing enabled but not required
| smb2-time:
|   date: 2026-03-24T12:28:12
|_  start_date: N/A
|_nbstat: NetBIOS name: DESKTOP-HBCRUDP, NetBIOS user: <unknown>, NetBIOS MAC: 22:a2:11:f7:34:62 (unknown)

TRACEROUTE
HOP RTT      ADDRESS
1   67.03 ms 192.168.100.107

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 39.60 seconds
yitelpin@anoen-x:~$
```

Analysis Questions

1. How many hosts were detected?

Number of hosts detected: 22

2. Which ports are most common?

The most common open ports observed in the network are:

- 135, 139, and 445 (TCP), commonly used by Windows systems for RPC, NetBIOS, and SMB services.
- Some devices also expose 49152 (dynamic port) and 62078 (iPhone synchronization service).

Most other hosts had all ports closed.

3. Why are open ports a risk?

Open ports mean services are listening and potentially accessible.

Risks include:

- Unauthorized access
  Attackers may connect to exposed services (e.g., databases, APIs)
- Vulnerability exploitation
  If a service has a known bug, it can be exploited remotely
- Information leakage
  Services may reveal:
    - software versions
    - system behavior
    - internal network details
- Entry points for attacks
  Each open port is a potential gateway into the system
- Brute-force attacks
  Services like SSH, databases, or APIs can be attacked repeatedly

4. What information can attackers gather?

From open ports and services, attackers can extract:

System information like

Operating system type and version
Running services (e.g., Docker, PostgreSQL, web server)

Service details like

Software versions (important for known vulnerabilities)
Configuration weaknesses

Network structure like

Internal IP ranges
Active hosts and services
Network discovery services (like WS-Discovery, mDN S)

Entry points like

Ports eg :

- 3000 → web application
- 5432 → database
- 22 (if open) → SSH

Behavioral patterns like

How services respond to requests
Timing, responses, and error messages